
Postroute Timing and DRV Fix with Manual ECO

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Purpose

Although EDA tools have become more and more powerful, as design nodes are being pushed towards extreme, there are still a few setup and hold timing violations and Design Rule Violations (DRV) left after the automatic repair by Innovus. To provide a handy and efficient solution and clear and straightforward fix steps, this document helps in manual ECO timing fix.

Audience

This document is intended for digital design engineers who need to fix timing violations manually at the post_route stage or want to learn the hands-on post_route timing fix methods.

Overview

This application note introduces methods to manually fix setup and hold timing violations and DRVs left by opt_design - post_route. Each violation is analyzed from three steps: report, fix, and verify. Related Innovus commands and applications are also included.

Stylus style is used in the application note.

Setup Violations

Report setup timing violations

To report setup timing violations, use the following command:

```
# Set report_timing check_type to setup.
set_db timing_analysis_check_type setup
# Set the fields to be reported by report_timing.
set_db timing_report_fields {timing_point load cell fanout
transition delay arrival}
# Report the negative slack timing path and include net as the
timing_point.
report_timing -net -max_slack 0 -max_paths 1000 \
>> timing_setup.rpt
```

Report Example:

Path1: VIOLATED (-0.117 ns) Late Output Delay Assertion

View func_ssg_0p81v_0c_cworst

Group: VCLK

Start point: (R) top/i1/reg_1/CK

Clock: (R) CLKIN

Endpoint: (R) add_o[23]

Clock: (R) VCLK

	Capture	Launch
Clock Edge: +	1.110	0.000
Src Latency: +	0.000	-0.020
Net Latency: +	0.600(I)	0.659(P)
Arrival: =	1.710	0.639
Output Delay: -	0.500	
Uncertainty: -	0.050	
Copr Adjust: +	0.000	
Required Time: =	1.161	
Launch Clock: =	0.639	
Data Path: +	0.639	
Slack: =	-0.117	

#Timing Point	Arc	Load (pf)	Cell	Fanout	Trans (ns)	Delay (ns)	Arrival (ns)
top/i1/reg_1/CK	CK	0.015	(arrival)	11	0.025	-	0.639
top/i1/wb_reg_1/Q	CK->Q	0.018	SDFEQ_C35	8	0.025	0.112	0.751
top/i1/wb	-	0.018	(net)	8	-	-	-
top/i1/g1/Y	AN->Y	0.006	NOR2B_C35	1	0.030	0.04	0.794
top/i1/n_355	-	0.006	(net)	-	-	-	-
top/i1/p115/Y	B->Y	0.009	NAND2_C35	4	0.018	0.019	0.813
top/i1/n_356	-	0.00	(net)	4	-	-	-

...

...

Path2: VIOLATED ...

Fix Setup Timing Violations

The setup timing fix methods are flexible. Adding repeaters on signal nets, resizing cells, and swapping cells are three typical and effective ways.

When a series of violations need to be fixed, it will be more efficient to legalize the added buffers at one go after all the fixes are done. Normally, auto refine place and update timing will be turned off by:

```
set_db eco_refine_place false
set_db eco_update_timing false
```

Then, the following commands are used in Innovus to fix the setup timing violations:

- Add repeaters:

```
eco_add_repeater -cells BUFX12 -net top/i1/ws \
  -location { x y }
```

Note: Normally, the buffer is inserted along the routing of the long net for setup timing fix.

- Resize cells:

```
eco_update_cell -instances top/i1/g1 -upsize
```

Note: Upsize the specified instance with the next available cell in the same footprint, or resize the cell directly to a much larger size.

- Swap cells:

```
eco_update_cell -instances top/i1/n_155 -cells BUFX16LVT
```

Note: Swap the HVT cell with LVT.

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Fix Step Examples:

- Add repeaters on signal nets

1. Read the timing report and find the relative large net delay in the violation path.

# Timing Point	Arc	Load (pf)	Cell	Fanout	Trans (ns)	Delay (ns)	Arrival (ns)
#							
u_cortexa7noram/u_ca7biu u write address buffers/waddr_source_reg_1_0/CK	CK	0.015	(arrival)	11	0.025	-	0.639
u_cortexa7noram/u_ca7biu u write address buffers/waddr_source_reg_1_0/Q	CK->Q	0.018	SDFFQ_X4M_A12TS_C35	8	0.025	0.112	0.751
u_cortexa7noram/u_ca7biu u write address buffers/n_3414	-	0.018	(net)	8	-	-	-
u_cortexa7noram/u_ca7biu u write address buffers/p0013A6986/Y	AN->Y	0.006	NOR2B_X8M_A12TS_C35	1	0.030	0.043	0.794
u_cortexa7noram/u_ca7biu u write address buffers/n_15120	-	0.006	(net)	1	-	-	-
u_cortexa7noram/u_ca7biu u write address buffers/p000606985/Y	B->Y	0.009	NAND2_X8A_A12TS_C35	4	0.018	0.019	0.813
u_cortexa7noram/u_ca7biu u write address buffers/n_14067	-	0.009	(net)	4	-	-	-
u_cortexa7noram/u_ca7biu u write address buffers/n000706988/Y	A->Y	0.005	NOR2_X6A_A12TS_C35	1	0.016	0.014	0.827
u_cortexa7noram/u_ca7biu u write address buffers/n_14070	-	0.005	(net)	1	-	-	-
u_cortexa7noram/u_ca7biu u write address buffers/FE_RC_51_0/Y	BB->Y	0.006	AOI21_X8M_A12TS_C35	1	0.011	0.017	0.845
u_cortexa7noram/u_ca7biu u write address buffers/n_3555	-	0.006	(net)	1	-	-	-
u_cortexa7noram/u_ca7biu u write address buffers/n0000057817/Y	A->Y	0.015	AND2_X11M_A12TS_C35	2	0.017	0.035	0.879
u_cortexa7noram/u_ca7biu u write address buffers/n_6387	-	0.015	(net)	2	-	-	-
u_cortexa7noram/u_ca7biu u write address buffers/fcpt7673/Y	A->Y	0.112	INV_X16M_A12TS_C35	21	0.014	0.035	0.914
u_cortexa7noram/u_ca7biu u write address buffers/n_15960	-	0.112	(net)	21	-	-	-
u_cortexa7noram/u_ca7biu u write address buffers/FE_0CPN34050_n_15960/Y	A->Y	0.005	BUFH_X3M_A12TS_C35	2	0.004	0.056	0.970
u_cortexa7noram/u_ca7biu u write address buffers/FE_0CPN34050_n_15960	-	0.005	(net)	2	-	-	-
u_cortexa7noram/u_ca7biu u write address buffers/p0026A6703/Y	BB->Y	0.006	AOI22_X4M_A12TS_C35	1	0.016	0.026	0.996
u_cortexa7noram/u_ca7biu u write address buffers/n_13637	-	0.006	(net)	1	-	-	-

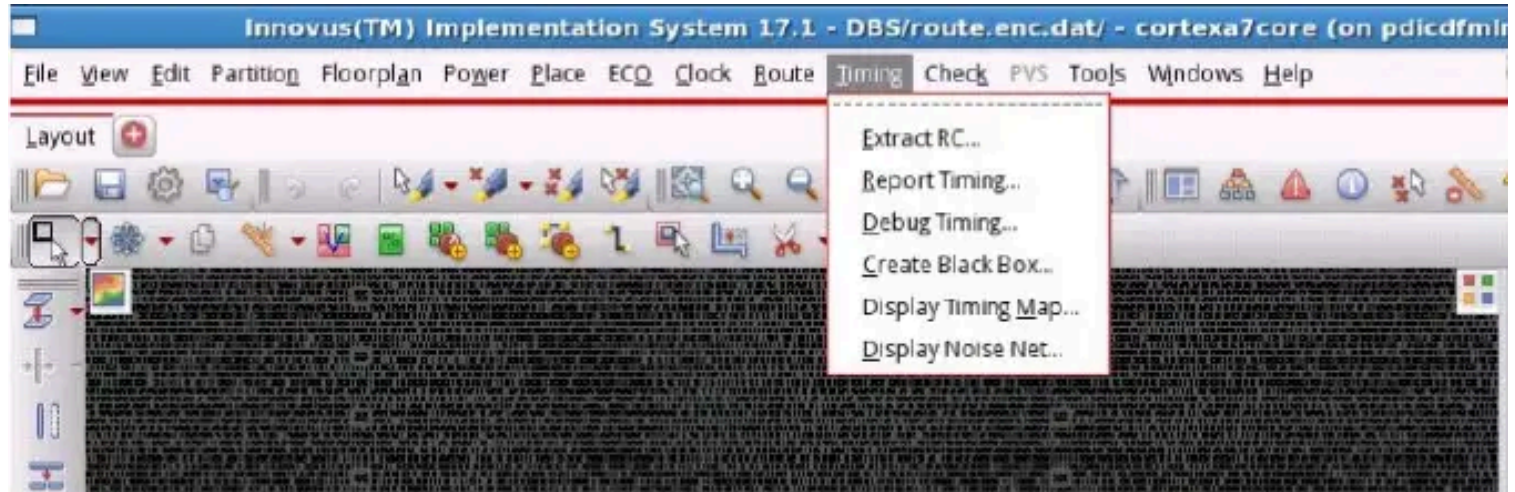
2. Use eco_add_repeater - cell s BUFH12 - net u_cortexa7noram/u_ca7biu u write address buffers/n_15960 \ - location {245. 64 455. 32} to insert the buffer.
3. Repeat steps 1-2 for other violation paths.
4. Call place_detail to legalize the added buffers.

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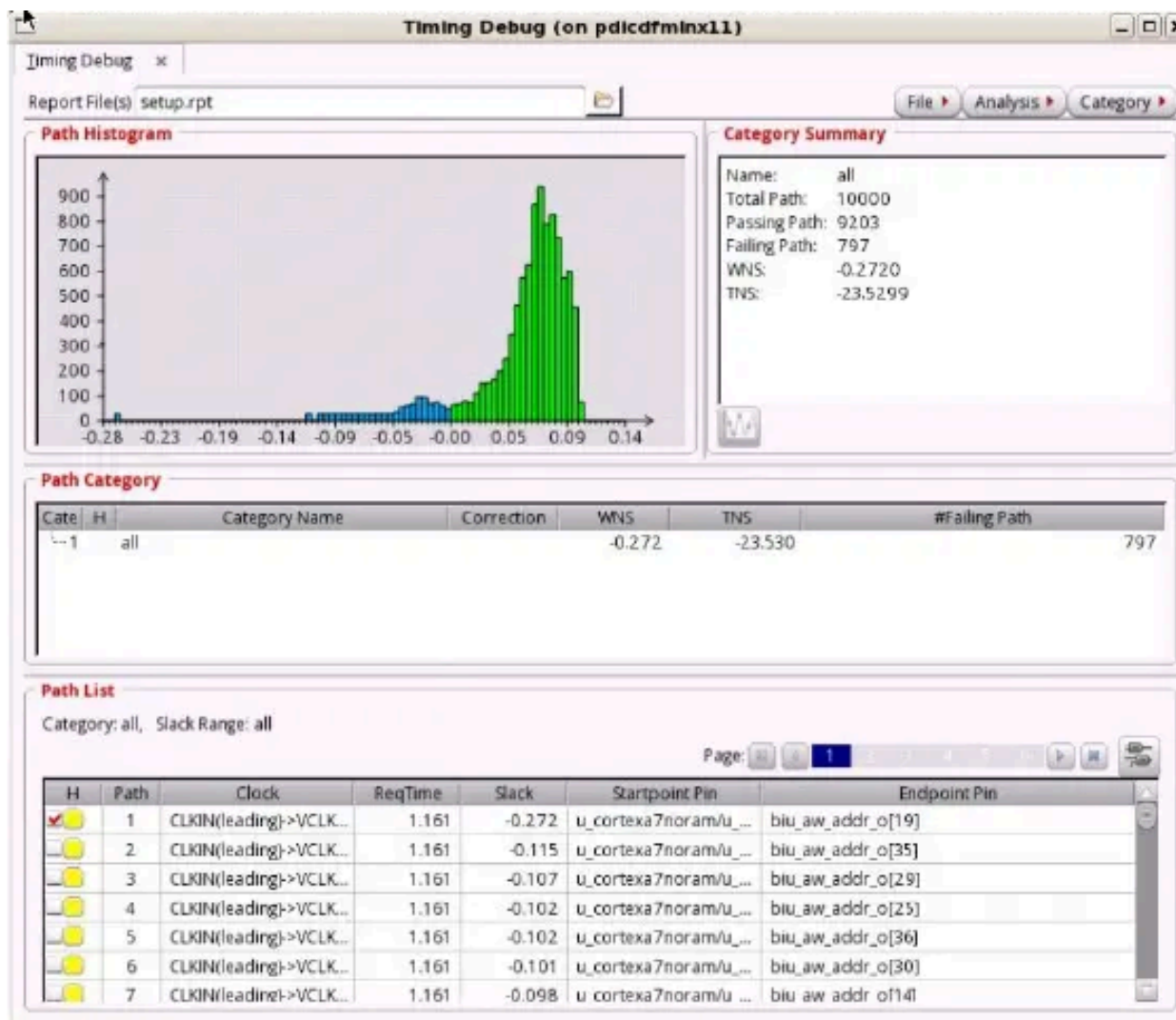
- Resize or swap cells

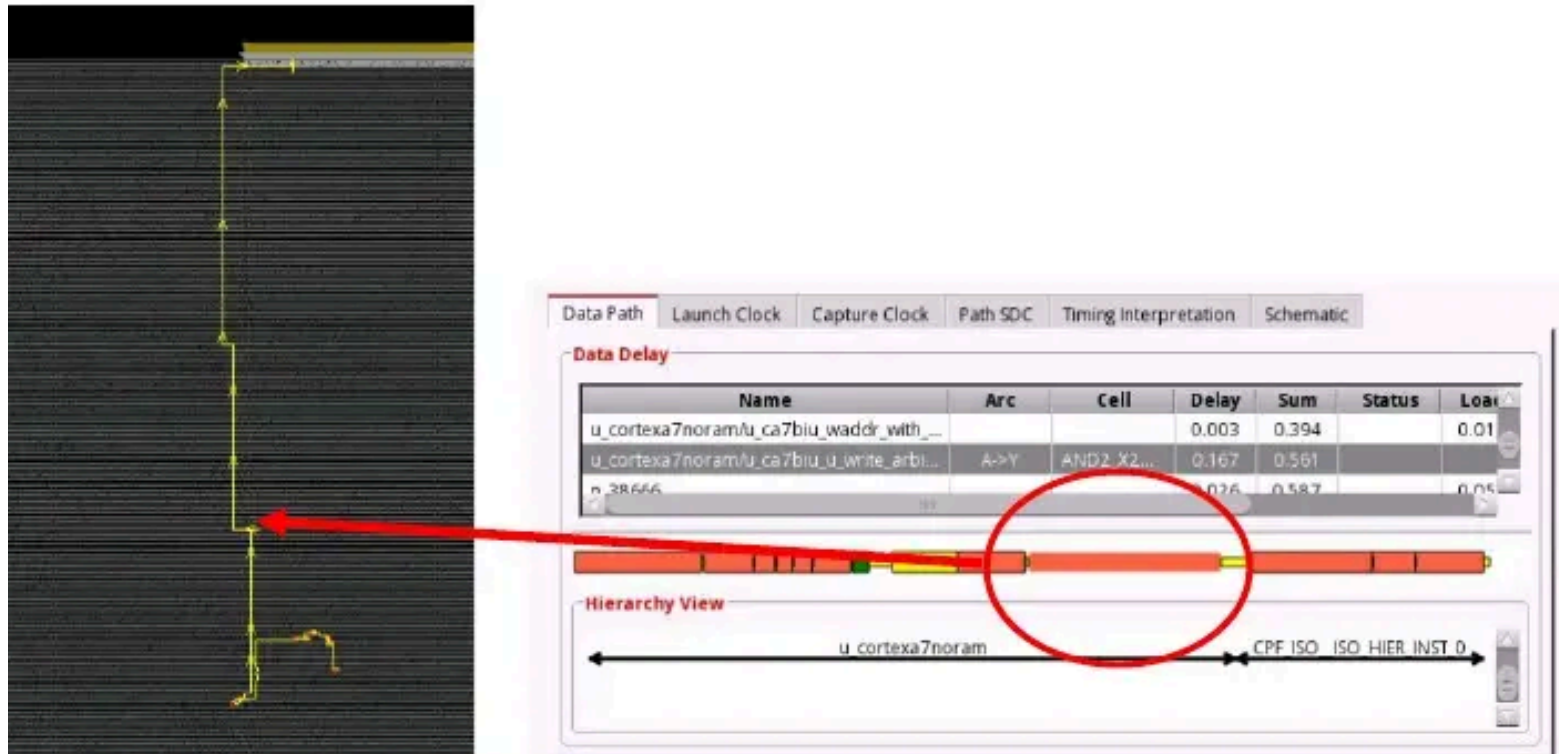
This method works when there are small size buffers or HVT cells in the violation path. Especially, this method will be highly efficient if the same cell appears in several violation paths.

1. Open the GUI and go to **Timing > Debug Timing**.



2. Select the first violation path. The path will be highlighted in the GUI.





3. Double-click the first row of the violation path. Timing Path Analyzer will pop up. The length of the rectangle represents the time of delay. As circled in red, the rectangle is an AND cell. The cell delay is almost 1/3rd of the whole path delay.

4. Apparently, this inst needs to be upsized or replaced.

```
eco_update_cell -insts u_cortexa7noram u_ca7biu_u_write_arbi / p1
\ -cells AND2_X11M_A12TS_C35
```

5. Repeat steps 1-4 for other violation paths.

6. Call place_detail to legalize the changed cells.

Verify Setup Timing Improvements

Call report_timing after the fixes. If violations remain, refer to the above methods to further fix until the timing report is clean.

Hold Violations

Report Hold Timing Violations

Like in the setup section, to report hold timing violations, use the following command:

```
#Set report_timing check type to hold.
set_db timing_analysis_check_type hold
#Set the fields to be reported by report_timing.
set_db timing_report_fields {timing_point load cell fanout
transition delay arrival}
#Report the timing path with negative slack and include net in the
timing_point.
report_timing -net -max_slack 0 -max_paths 1000 >> timing_hold.rpt
```

Report Example:

```
Path1: VIOLATED (-0.016 ns) Hold Check with Pin top/i1/reg_3/CK->D
      View func_ssg_0p81v_0c_cworst
      Group: CLKIN
      Start point: (F) cluster_i[3]
      Clock: (R) VCLK
      Endpoint: (F) top/i1/reg_3/D
      Clock: (R) CLKIN
```

	Capture	Launch
Clock Edge: +	0.000	0.000
Drv Adjust: +	0.000	0.024
Src Latency: +	-0.020	0.600
Net Latency: +	0.655(P)	0.000(I)
Arrival: =	0.635	0.624
Hold: -	-0.023	
Uncertainty: -	0.050	
Copr Adjust: +	0.000	
Required Time: =	0.661	
Launch Clock: =	0.624	
Data Path: +	0.021	
Slack: =	-0.016	

#Timing Point	Arc	Load (pf)	Cell	Fanout	Trans (ns)	Delay (ns)	Arrival (ns)
cluster_i[3]	cluster_i	0.052	(arrival)	1	0.076	0.021	0.645
cluster_i[3]	-	0.052	(net)	1	-	-	-
top/i1/reg_3/D	D	0.052	SDFEQ_X1M_C35	1	0.098	0.021	0.645

Path2: VIOLATED

Fix Hold Timing Violations

To fix hold timing violations, the same commands are used but the theory is opposite. Usually, `up_size` or HVT-to-LVT cell swapping is used to fix setup time. However, to fix hold timing, `down_size` or LVT-to-HVT cell swapping should be specified. Further, buffers should be inserted at the middle of the net to fix the setup timing violation. To fix hold timing, buffers need to be added close to the sink. Then, the following commands are used in Innovus to fix the hold timing violations:

- Add repeaters:

```
eco_add_repeater -cell BUFX12 -net n1 \
  -relative_distance_to_sink 0.1
```

Note: `-relative_distance_to_sink` only works with `-pins` or `-net` with 1-sink net on a routed design.

- Resize cells:

```
eco_update_cell -instance1 -down_size
```

Note: Downsize the specified instance with the next available cell in the same footprint, or resize the cell directly to a much smaller size.

- Swap cells:

```
eco_update_cell -instance2 -cell BUFX16HVT
```

Note: Swap the standard voltage threshold cell to a high value.

Fix Step Examples:

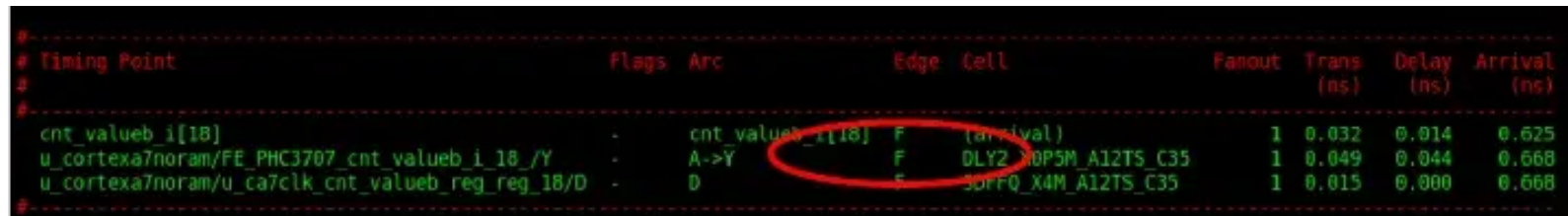
As in the setup timing fix, first, turn off auto refine place and update timing:

```
set_db eco_refine_place false
set_db eco_update_timing false
```

- Add repeaters on signal nets
 1. Find the violation path in the timing report
 2. Choose the net for buffer insertion.
 3. If it is a 1-sink net, use `eco_add_repeater \`
 `- relative_distance_to_sink 0.1 - net`
`cluster_i[3] \` `- cell DLY2_X3M_C35`

Note: Normally, “delay cell” is used to fix hold timing.

4. Repeat steps 1-3 for other violation paths.
 5. Call `place_detail` to legalize the added buffers.
- Resize or swap cells
 1. Find the violation path with cells that can be resized in the timing report.



Timing Point	Flags	Arc	Edge	Cell	Fanout	Trans (ns)	Delay (ns)	Arrival (ns)
cnt_valueb_i[18]	-	cnt_valueb_i[18]	F	(arrival)	1	0.032	0.014	0.625
u_cortexa7noram/FE_PHC3707_cnt_valueb_i_18/Y	-	A->Y	F	DLY2_X0P5M_A12TS_C35	1	0.049	0.044	0.668
u_cortexa7noram/u_ca7clk_cnt_valueb_reg_reg_18/D	-	D	F	DLY2_X0P5M_A12TS_C35	1	0.015	0.000	0.668

2. Use `eco_update_cell -insts \`
`u_cortexa7noram/FE_PHC3707_cnt_valueb_i_18_ -downsize`
3. Repeat steps 1-2 for other violation paths.
4. Call `place_detail` to legalize the changed cells.

Verify Hold Timing Improvements

Call `report_timing` after the fixes. If violations remain, refer to the above methods to further fix until the timing report is clean.

DRV

Report DRV

`max_tran`, `max_cap`, and `max_fanout` are three common DRVs after `opt_design`. `max_cap` and `max_fanout` violations are two factors that affect the max transition time. So, the `max_tran` violation is the key point of DRV fix.

```
#Enables opt_design to fix fanout load violations
set_db opt_fix_fanout_load true
#Lets the tool perform optimization before do manual fix
opt_design -post_route
```

The transition violation report is at `.timingReports/Top_postRoute.tran.gz`.

Report Example:

#Net / Inst Pin	MaxTranTime	TranTime	TranSlack	Cell Port	Remark
N_13013					
p001/ A	0.178r / 0.178f	0.211r / 0.215f	-0.033r / -0.037f	AOI_X1/ A	R
p002/ B	0.178r / 0.178f	0.211r / 0.215f	-0.033r / -0.037f	AOI_X2/ A	R
N_2580					
data/ D	0.178r / 0.178f	0.181r / 0.188f	-0.003r / -0.010f	RF1_SVT/ D	R
...					
...					

All manual fixes for DRVs are based on this report.

Fix DRV

Since both `max_cap` and `max_fanout` are influencing factors of `max_tran`, the `max_tran` violation fix is illustrated below. First, find the violation path specified in the first column in the report. Then, refer to the value in the TranSlack column to choose the size of the buffer being added. Normally, a medium size buffer such as X12 or X16 is a good choice to start with. Unlike setup and hold time fix, which have different ways for different conditions, the most widely used method for DRV fix is adding buffers.

```
eco_add_repeater -cells BUF_X12 -net n1 \
  -off_load_at_location {x y} | -off_load_slack ns
```

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- of f _l oad _a t _l ocat i o n or - of f _l oad _s l a c k is typically used for DRV fix. The former is more common. Both options target on reducing the load on the net driver.

- of f _l oad _a t _l ocat i o n allows buffer insertion at a specified location on the violation path. The tool will determine the buffer direction automatically.

- of f _l oad _s l a c k inserts a buffer to offload paths with less than the specified slack threshold. - net specified with - of f _l oad _s l a c k is not the violation path. It can be the net before the violation path and must have enough positive slack to sacrifice. It improves the critical path timing by penalizing noncritical path delays.

Like the former sections, turn off auto refine place and update timing before starting to fix:

```
set_db eco_refine_place false  
set_db eco_update_timing false
```

Fix Step Example:

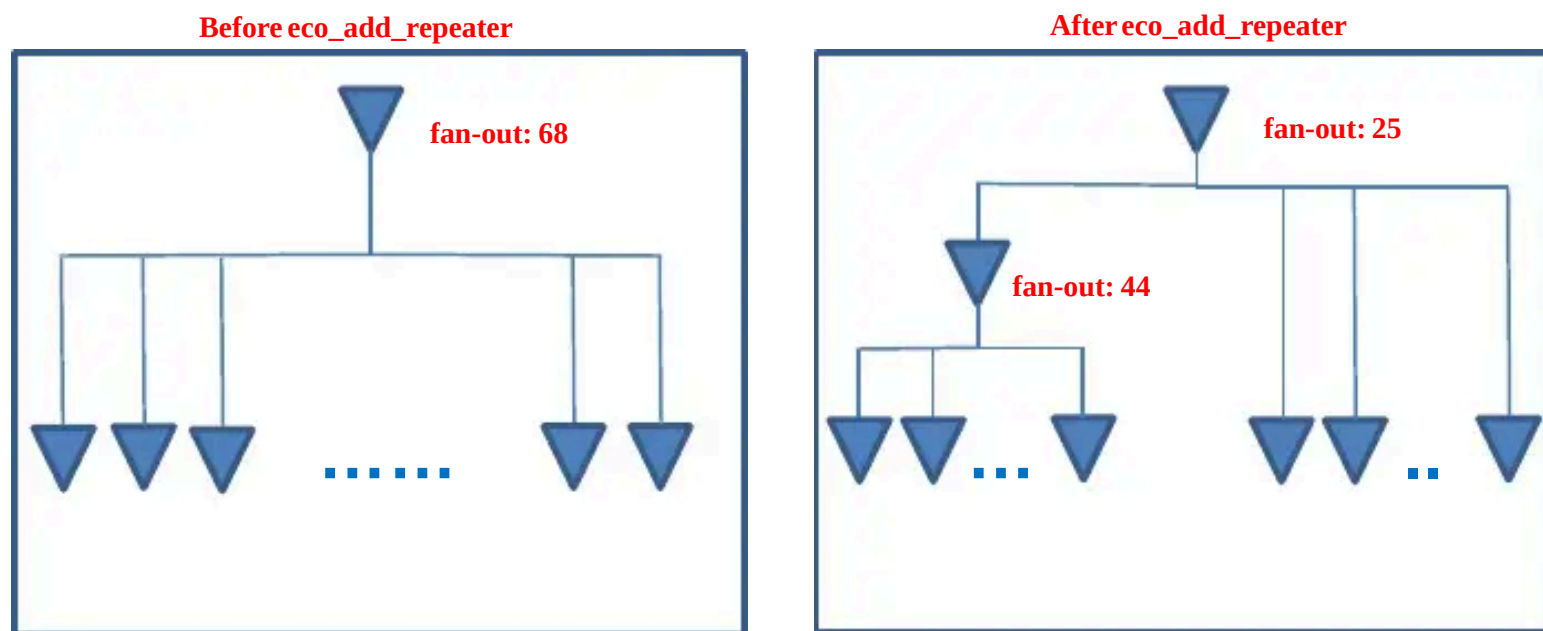
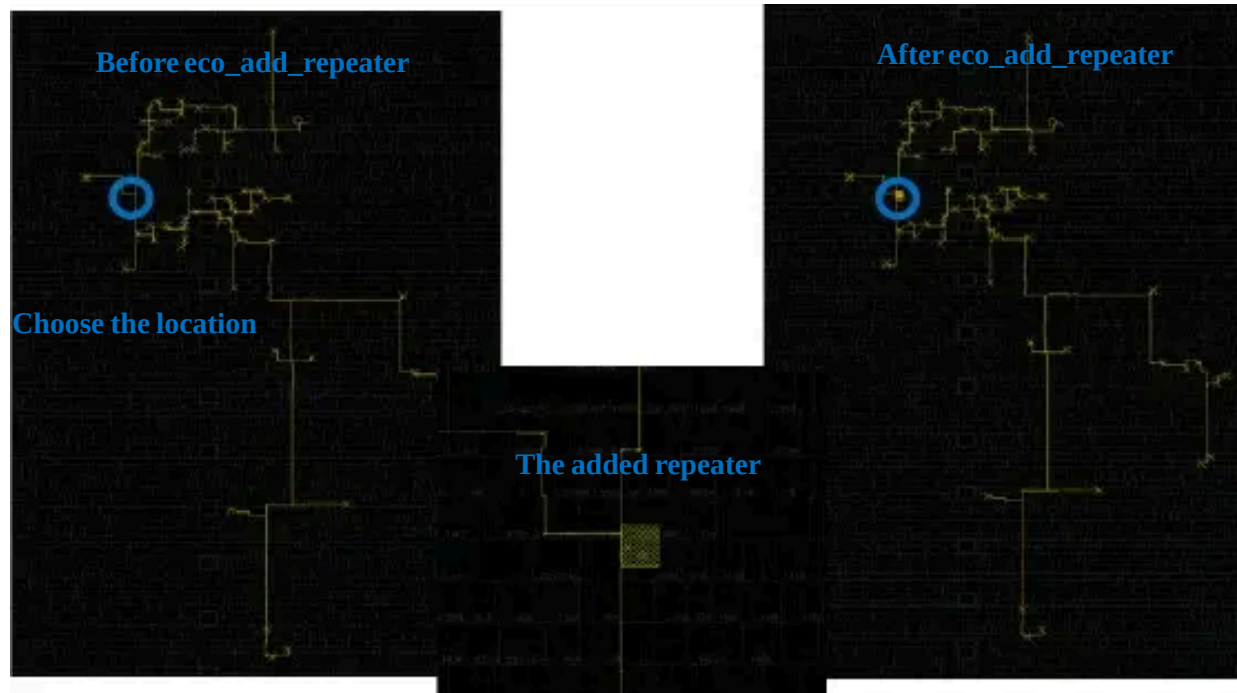
1. Find the violation path with all its sinks in the GUI.



2. Decide the location for buffer insertion. Normally, a point that can offload more sinks is better.

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3. Use `eco_add_repeater -off_load_at_location {380.4 452.8} \`
- cell `BUFH_X4M_A12TS_C35` - net `DFTSE_FE_COEN77`
to drive part of sinks.



4. Repeat steps 1-3 for all violation paths.
5. Call `place_detail` to legalize the added buffers.

Innovus can add buffers while keeping the original routing without causing any detour. This makes Innovus a more intelligent and reliable tool for ECO operation.

Verify DRV clean

Call `time_design - post_route` after buffer insertions and then check the output reports. If `max_tran` violations remain, refer to the above methods to further fix until the DRV is clean. Normally, when the `max_tran` violation is clean, `max_cap` and `max_fanout` violations are clean as well.

Summary

Although manual ECO timing fix is still inevitable in the digital design, it is good that methods for the timing fix do not change much as the design node does. With skills for manual timing fix, designers can handle most of the designs, irrespective of the design technology.

This application note illustrated a series of powerful and useful solutions for setup and hold violations and DRVs at the `post_route` stage.

References

- Innovus Stylus Common UI Text Reference Manual
 - Database Access and Handling Commands
 - Interactive ECO Commands and Attributes
 - Timing Analysis Commands and Attributes
 - Timing Optimization Commands and Attributes

Support

Cadence Support Portal provides access to support resources, including an extensive knowledge base, access to software updates for Cadence products, and the ability to interact with Cadence Customer Support. Visit <https://support.cadence.com>.

Feedback

Email comments, questions, and suggestions to content_feedback@cadence.com.

Appendix

Stylus and Legacy Command Mapping

Stylus	Legacy
set_db_timing_analysis_check_type	setAnalysisMode - checkType
set_db_timing_report_fields	set_global_report_timing_format
place_detail	refinePlace
report_timing	report_timing
set_db_eco_refine_place	setEcoMode - refinePlace
set_db_eco_update_timing	setEcoMode - updateTiming
eco_add_repeater	ecoAddRepeater
eco_update_cell	ecoChangeCell
set_db_opt_fix_fanout_load	setOptMode - fixFanoutLoad
opt_design	optDesign
time_design	timeDesign